Case 04: Steady Transport and Immediate Ignition

1 Verification purpose

This case verifies the limiting mechanism shared by steady Eulerian transport and zero-delay wildland ignition. The isolated transport simulation condition disables surface spread and uses an areal source; wildland ignition without delay uses a fire-power-based source and additionally checks accumulated population.

The paired design exposes failures in generation, transport, deposition, direct ignition, and time-of-arrival (TOA) updates. Success requires both simulation conditions to recover wind-speed arrival behavior and the wildland simulation condition to recover its one-metre-strip accumulation reference.

2 Mathematical formulation and numerical implementation

For constant wind speed $U = 6.71$ m/s and zero ignition delay, an ember-deposited cell becomes an active source immediately. Translation invariance then gives

$$t_{\text{ref}}(x) = \frac{x - x_0}{U}, \qquad x_{\text{LE,ref}}(t) = x_{\text{LE},0} + U(t - t_0), \qquad \overline{ROS}_{\text{ref}} = U, \tag{1}$$

where x_0 is the first usable cell centre. The final arrival-time raster stores the complete cell-crossing history used for the time of arrival (TOA) profile metric. Leading-edge history is evaluated independently from every dumped instantaneous level-set field. At the physical centre row, the postprocessor linearly interpolates every adjacent pair of cell-centred ϕ values that changes sign and retains the rightmost $\phi = 0$ crossing. Here $x_{\text{LE},0}$ is the first extracted crossing at dump time t_0. Once the zero contour has left the physical domain, that dump has no observable in-domain leading edge and is excluded from the trajectory fit rather than being replaced by a stationary domain-edge value. Mean rate of spread (ROS) is the least-squares slope through all observable dump samples.

For immediate-ignition simulation condition, a burning cell emits during its surface-fire residence time. Using the FLIN and spread rate written by the same executed cell gives

$$t_{\text{emission}} = \frac{\Delta x}{VS}, \qquad N_{\text{ref}} = GR_{1m}\,\frac{FLIN}{1000}\,\frac{\Delta x}{VS}. \tag{2}$$

Here FLIN is in kW m^{-1} and VS is converted from ft min^{-1} to m s^{-1}. The documented nominal FLIN and VS give 1497.2 particles for this illustrative state [1]; the metric uses run-derived values so it remains consistent with the active fuel calculation. ELMFIRE forms pixel fire power from fireline intensity times cross-stream width Δy. The configured coefficient $GR_{\text{ELM}} = 33.3/\Delta y$ pcs/s/MW therefore makes the raw pixel count represent a one-metre strip. For the one-dimensional unit-width profile, each centre-row pixel count is converted to the density of the represented one-metre strip,

$$n_i = \frac{N_i}{\Delta x(1\text{ m})} \quad [\text{pcs m}^{-2}]. \tag{3}$$

The two-cell numerical halo is excluded. The complete physical raster is still inspected, and its all-row deposited total is retained as a conservation diagnostic, but it is not mixed into the one-dimensional plotted observable. For the PER-AREA simulation condition, the configured source gives $n_\infty = GR\Delta x\tau_{\text{ember}}/(1\text{ m})$. For the PER-MW simulation condition, Equation 2 divided by $\Delta x(1\text{ m})$ gives the run-derived density reference. A zero or nodata final burned-cell FLIN or VS value does not define an independent residence-time reference. The evaluator retains only finite cells with $FLIN > 0$ and

$VS > 0$, requires at least two such cells in the declared plateau, and otherwise reports the accumulation component as **NOT EVALUABLE** rather than forming a relative error against zero.

The discrete algorithm is: generate embers in every active source cell; distribute them among downwind bins; accumulate them in landing cells; apply DIRECT ignition with $P_{ign} = 1$ and zero local/cell delay; update ember-ignition, level-set, and TOA states together; then advance the level-set clock. The two simulation conditions differ only where required by their source experiment.

3 Derivation of expected behavior

A deposition distribution with finite support changes which upstream cells contribute near x_0, but not the propagation characteristic in Equation 1. Consequently TOA should be linear, and the leading edge extracted from ϕ should oscillate by at most one cell around $x_{\mathrm{LE},0} + U(t - t_0)$. Its fitted slope should approach U. Every cell marked by the ember-ignition raster must have finite nonnegative TOA.

In both simulation conditions, locations farther than the maximum travel distance receive contributions from a complete set of upstream emitters. Conservation then produces the configuration-consistent plateaus. The PER-AREA source uses $GR = 10$ pcs m^{-2} s^{-1}, $\Delta x = 10$ m, and $\tau_{\text{ember}} = 6$ s, giving 600 pcs m^{-2} under the represented-strip normalization. The PER-MW reference is calculated from co-located final FLIN and VS. The 155–195 m interval contains cells with complete upstream support.

The wind Courant–Friedrichs–Lewy (CFL) upper-bound step is

$$\Delta t_{\text{max}} = 0.5(10)/6.71 = 0.745156 \text{ s}.$$

To retain the requested 100-s stop as an exact multiple, preprocessing selects $N = \lceil 100/0.745156 \rceil = 135$ and $\Delta t = 100/135 = 0.740741$ s. The resulting wind CFL is 0.4970, below the declared 0.5 bound. With $x_0 = 5$ m, representative exact TOA substitutions in the declared plateau interval are

$$t_{\text{ref}}(155) = \frac{155 - 5}{6.71} = 22.3547 \text{ s}, \qquad t_{\text{ref}}(195) = \frac{195 - 5}{6.71} = 28.3159 \text{ s}.$$

Likewise $x_{\mathrm{LE,ref}}(50) = 5 + 6.71(50) = 340.5$ m, and the front reaches the 500-m boundary after $(500 - 5)/6.71 = 73.7705$ s. These substitutions give exact reference values of zero for TOA relative error, leading-edge NRMSE, and ROS relative error before raster quantization.

For the PER-AREA member, the complete-support density is

$$n''_{\infty} = GR\,\Delta x\,\tau_{\text{ember}}/(1 \text{ m}) = 10(10)(6) = 600 \text{ pcs m}^{-2}.$$

For the PER-MW member, the reference evaluated at each co-located FLIN/VS cell is

$$n''_{\text{ref}} = 33.3\frac{FLIN/1000}{VS(0.3048/60)} = 6.555118\frac{FLIN}{VS} \text{ pcs m}^{-2},$$

with FLIN in kW m^{-1} and VS in ft min^{-1}. The historical nominal substitution gives 1497.2 particles; current acceptance uses the generated co-located fields rather than that rounded historical value. Both plateau references therefore have expected relative error zero and are tested with a 0.10 allowance.

4 Verification metrics and acceptance criteria

For valid centre-row cells, define $e_i = |t_i - t_{\text{ref,i}}|/t_{\text{ref,i}}$. Each simulation condition requires at least two cells, $\bar{e} \le 0.10$, and $\max e_i \le 0.25$. Leading-edge error is the RMSE relative to $x_{\mathrm{LE},0} + U(t - t_0)$ divided by $U(t_{\text{last}} - t_0)$ and must not exceed 0.10; t_{last} is the last dump with an in-domain $\phi = 0$ crossing. Every scheduled time must have exactly one matching ϕ raster. Fitted mean-ROS relative error must

not exceed 0.10. All ember-ignited physical cells must have finite, nonnegative TOA. Each simulation condition additionally requires $|\widetilde{n} - n_{\text{ref}}|/n_{\text{ref}} \leq 0.10$, where $\widetilde{n}$ is the median centre-row density from Equation 3 over the fully developed 155–195 m plateau. The corresponding analytical or run-derived reference is calculated independently of the ember-flux values. For the PER-MW simulation condition, the same mask is applied to the observed density and the co-located FLIN/VS reference, with at least two finite, strictly positive reference cells required. Insufficient support makes the metric NOT EVALUABLE; a zero denominator is never treated as a failed relative-error comparison. Domain-wide deposited totals are also recorded as diagnostics. Exactly two prescribed simulation conditions and one final raster of every required kind are mandatory. The overall decision is the conjunction of completeness and all component decisions; missing or stale products are non-passing.

4.1 Justification of metric quantification

The 10% limits on mean TOA, leading-edge NRMSE, fitted ROS, and plateau density are engineering verification bands for a first-order, cell-centred transport/ignition calculation. They are broad enough to accommodate grid-crossing time quantization and the use of a localized run-derived FLIN/VS reference, yet they remain small compared with the order-one signature of a nonpropagating or solver-step-limited implementation. The 25% pointwise TOA limit is intentionally looser than the mean limit because a single cell next to the source or front can carry a disproportionately large relative timing error; requiring at least two cells and the 10% mean prevents that allowance from masking a systematic bias. The plateau median is used because the expectation is spatially constant and the median is insensitive to one raster edge cell. Ignition-state consistency has zero physical tolerance: a cell cannot be classified as ember ignited without a finite, nonnegative arrival time. Both simulation conditions and all components are conjunctive because transport, ignition, and accumulation are separate claims of this combined test.

4.2 Predeclared acceptance criteria

Table 1 summarizes the metrics fixed before inspection of the calculated results. Numerical values and component statuses are reported separately in Section 7.

Table 1: Predeclared verification metrics and acceptance criteria.

Metric	Output, region, and aggregation	Acceptance criterion
Output completeness	Two prescribed simulation conditions with one final TOA, ignition-state, and accumulated-firebrand product of each required kind, plus one ϕ raster for every scheduled dump time.	2 of 2 simulation conditions complete; no missing ϕ dumps
TOA profile error	Mean and maximum of $\lvert t_i - t_{ref,i}\rvert / t_{ref,i}$ on valid centre-row physical cells.	Mean $\leq$ 0.10; maximum $\leq$ 0.25
Leading-edge error	RMSE of the rightmost interpolated $\phi = 0$ crossing relative to $x_{LE,0} + U(t - t_0)$, normalized by the observable reference displacement.	NRMSE $\leq$ 0.10
Mean-ROS error	Relative error of fitted leading-edge ROS against $U = 6.71\,\mathrm{m\,s^{-1}}$.	$\leq$ 0.10
Ignition-state consistency	Every ember-ignited physical cell has a finite, nonnegative TOA.	No inconsistent cells
Accumulation reference support	Finite positive co-located FLIN/VS cells over 155–195 m; zeros and nodata cannot define the PER-MW residence-time reference.	At least two cells; otherwise NOT EVALUABLE
Accumulation plateau error	Relative error of median one-metre-strip density over 155–195 m against the independent reference, using the same valid-reference mask for the PER-MW simulation condition.	$\leq$ 0.10 for both simulation conditions
Overall decision	Conjunction of completeness and every component for both simulation conditions.	PASS only if all pass

5 Simulation configuration and predicted outcome

Figure 1 records the prepared spatial inputs for isolated transport. These panels show the prepared spatial fields, remove the two-cell numerical buffer, and retain map coordinates in metres. The left panel shows the most informative fuel or structure layout available for the case; the right panel shows the cells selected by the non-positive initial level-set field, making a localized ignition or firebrand-emission source explicit.

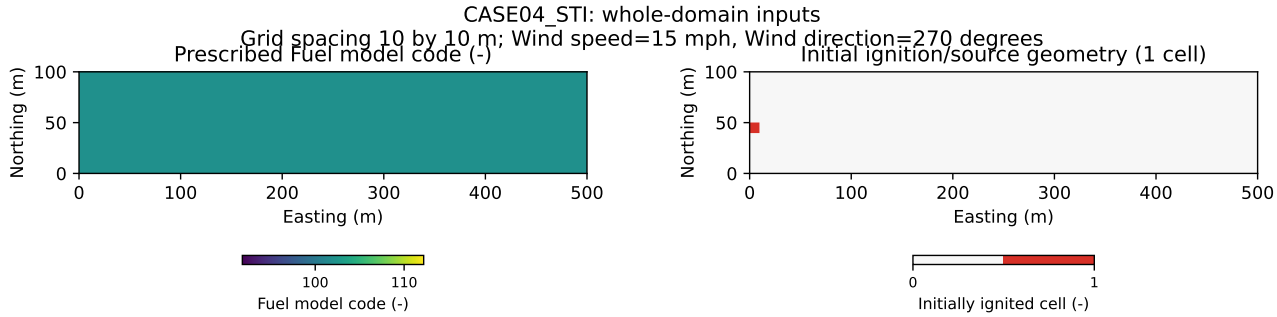


Figure 1: Whole-domain prepared input configuration for the representative simulation condition. The physical domain is shown without the numerical halo; coordinates, configured wind values, fuel or structure layout, and initial ignition/source geometry are identified in the panels.

Parameter	Configured value	Units	Verification role
Domain	500×100	m	common physical extent
Grid and halo	10; two cells/side	m	prescribed resolution; excluded halo
Time step	$100/135 = 0.740741$	s	wind CFL $0.4970 < 0.5$
Wind	6.71, from 270°	m/s	constant transport velocity
Fuel/terrain	fire behavior fuel model (FBFM) 102; flat	–	uniform wildland control
Transport simulation condition	PER-AREA; surface off; 100 s	–	isolates transport-impulse simulation condition transport
Wildland simulation condition	PER-MW; surface on; 100 s	–	immediate-ignition simulation condition residence source
Transport/ignition	EMPIRICAL; EULERIAN; DIRECT	–	current spotting path
Ignition response	$P_{ign} = 1$; delays zero	–	immediate limiting regime
Crosswind/consumption	disabled/disabled	–	one-dimensional conservation

Table 2: Configuration shared by and distinguishing the two combined simulation conditions.

Before inspecting output, both TOA curves should overlay $(x - x_0)/U$. Both dumped-ϕ leading-edge histories should follow $x_{\mathrm{LE},0} + U(t - t_0)$ with grid-scale steps, and their mean ROS should be near 6.71 m/s. The two density bar profiles should rise near the inlet and approach their horizontal references: 600 pcs m^{-2} for the PER-AREA transport simulation condition and the FLIN/VS-derived value for the wildland simulation condition.

6 Actual simulation results

Figure 2 shows the representative accumulated firebrand density from isolated transport. The displayed field is taken from the simulated results, masks missing values, and excludes the two-cell numerical buffer. Firebrand counts are normalized by the cell length and the represented 1-m cross-stream strip, giving units of pcs m^{-2}.

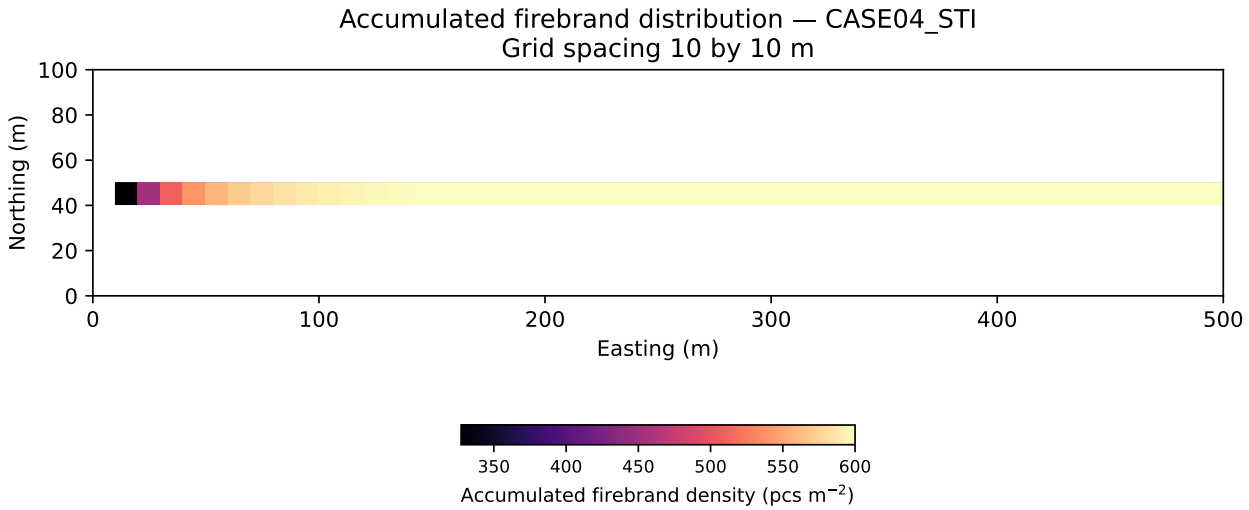


Figure 2: Whole-domain accumulated firebrand density from the representative completed ELMFIRE run. This domain-scale result supplements, rather than replaces, the higher-level verification profiles, convergence plots, ensemble statistics, and calculated acceptance metrics below.

The runner regenerated both input stacks and postprocessed only the current ELMFIRE products recorded in prescribed simulation conditions. Figure 3 presents the four physical observables used in the steady-transport reference experiment: accumulated density, arrival time, leading-edge position, and instantaneous leading-edge rate of spread.

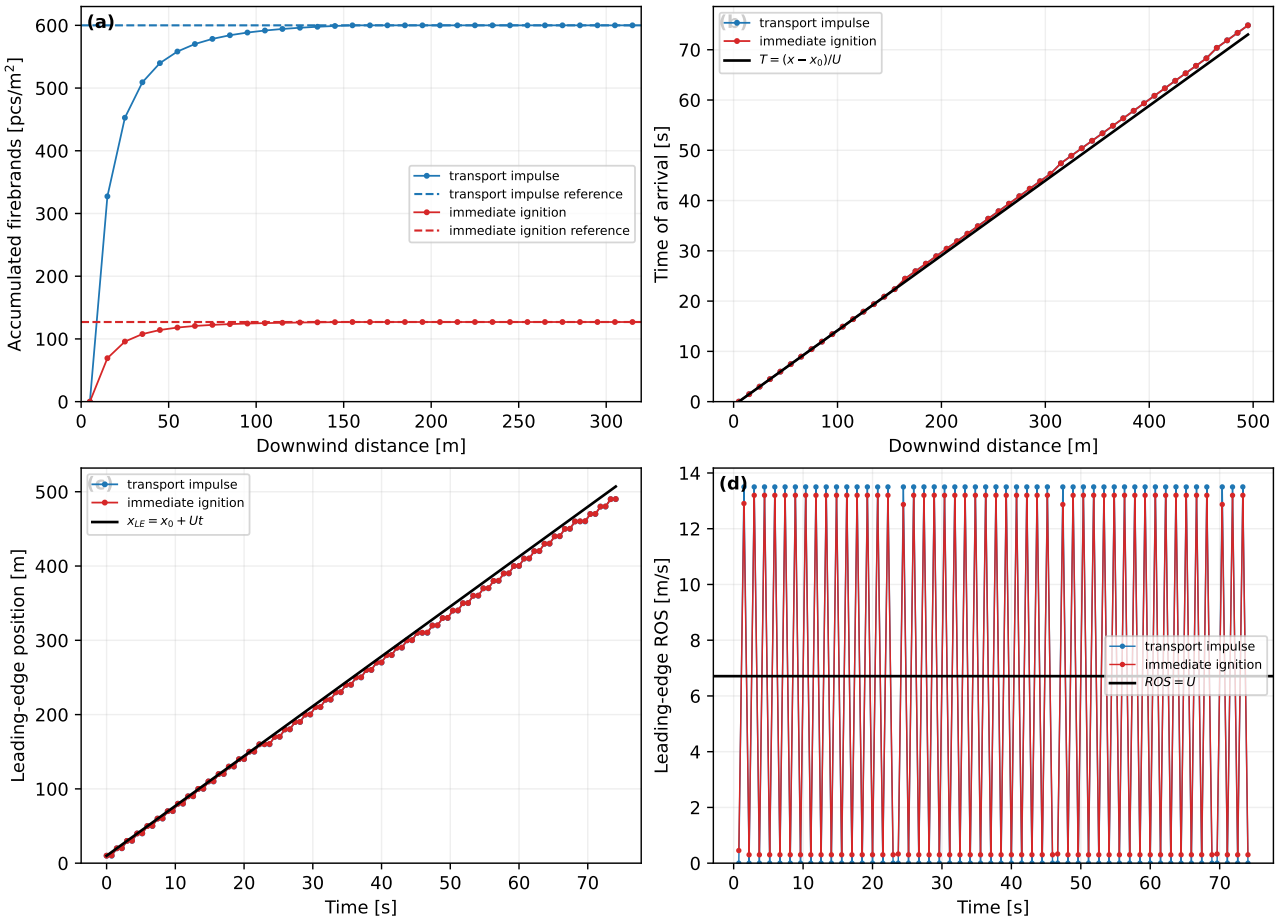


Figure 3: Steady-transport and immediate-ignition observables from actual ELMFIRE outputs. Solid colored curves identify the two controlled simulation conditions; black or color-matched reference curves show the derived analytical behavior.

7 Calculated metrics and verification decision

Metric	Limit	Calculated	Status
Output completeness	2 simulation conditions	2/2	PASS
Transport ϕ history	all scheduled dumps	137/137 dumps; 101 contours	PASS
Transport TOA mean error	≤ 0.10	0.01476	PASS
Transport leading-edge NRMSE	≤ 0.10	0.0165	PASS
Transport mean-ROS error	≤ 0.10	0.02537	PASS
Wildland ϕ history	all scheduled dumps	137/137 dumps; 101 contours	PASS
Wildland TOA mean error	≤ 0.10	0.01476	PASS
Wildland leading-edge NRMSE	≤ 0.10	0.01639	PASS
Wildland mean-ROS error	≤ 0.10	0.02536	PASS
Transport accumulation error	≤ 0.10	8.138e-08	PASS
Wildland accumulation error	≤ 0.10	5.625e-08	PASS
Transport deposited total	diagnostic	2.869e+05 pcs	–
Wildland deposited total	diagnostic	2.099e+05 pcs	–
Overall decision	all components pass	PASS	**PASS**

Table 3: Calculated verification metrics for the completed simulations.

8 Technical interpretation

The combined decision distinguishes a transport defect from a source-duration or normalization defect. Agreement in transport-impulse simulation condition but not immediate-ignition simulation condition implicates the residence-time PER-MW path; disagreement in both implicates shared transport, direct ignition, or TOA state handling. A full-area total of 2.099e+05 particles was deposited across 5 active rows. A plateau-only failure cannot be hidden by a plausible TOA curve, and an average ROS cannot hide local TOA or product inconsistency. The numerical values in the table and figure provide the executed source revision's decision rather than a qualitative visual judgment.

9 Scope and limitations

This is deterministic numerical verification of steady one-dimensional Eulerian transport and immediate ignition. It does not validate the empirical spotting PDF against experiments, time-varying wind, crosswind transport, stochastic ignition, delayed ignition, Lagrangian tracking, or consumption. Surface spread is present in the immediate-ignition simulation condition source calculation but the wind-speed leading edge is expected to dominate.

References

[1] Y. Qin, *A Physics-Based Eulerian Framework for Modeling Firebrand Showering in Regional-Scale Wildland and Wildland–Urban Interface Fire Simulations*, Ph.D. dissertation, University of Maryland, College Park, 2025.